

The empirical recurrence for OEIS A238173, proved by transfer matrix

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Abstract

OEIS A238173 counts arrays over $\{0, \dots, 2\}$ subject to a condition on the order statistics of every 2×2 subblock, and carries an empirical recurrence of order 6 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The condition is local to two consecutive rows, so the count is a walk count on $S = 19$ states and is C -finite; whether the conjectured recurrence annihilates it is then settled by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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1 The conjecture

OEIS A238173 is “Number of $(n+1) \times (1+1)$ 0..2 arrays with the maximum plus the minimum of every 2×2 subblock equal.”. It has offset 1 and begins

81, 423, 2329, 13447, 79641, 479703, 2918233, 17859079, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = 10*a(n-1) - 22*a(n-2) - 20*a(n-3) + 63*a(n-4) - 6*a(n-5) - 24*a(n-6)$.

As of the “Last modified” line on the live entry (Oct 24 2018 08:21:40, revision 7) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

Sort the four entries of a 2×2 subblock as $a \leq b \leq c \leq d$: they are its minimum, lower median, upper median and maximum. The entry fixes the signed sum

$$\sigma = a + d,$$

and asks that σ take the SAME value on every 2×2 subblock of an $(n+1) \times 2$ array over $\{0, \dots, 2\}$.

The common value is not named, so the count splits by it. For a fixed v let $a_v(n)$ count the arrays all of whose subblocks give $\sigma = v$. An array determines its common value, so the sets belonging to different v are disjoint and $a(n) = \sum_v a_v(n)$, a finite sum.

3 The count is a walk count

For a fixed v the condition involves two consecutive rows and nothing else: rows r and s may follow one another exactly when σ equals v on each of the 1 subblocks they form. So a row is a state, an $(n+1)$ -row array is a walk of n steps, and with M the adjacency matrix on the disjoint union over v ,

$$a(n) = \iota^\top M^n \tau, \quad \iota = \tau = (1, \dots, 1)^\top.$$

Rows carrying no edge at all are dropped: such a row can only be the whole of a one-row array, which is not among the objects counted. There are $S = 19$ states left. The alignment of the walk index with the entry's own n was checked against its published terms rather than assumed. In particular a is C -finite.

4 The proof

Lemma 1. *Let $q(t) = t^6 - \sum_i c_i t^{6-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+6} - \sum_i c_i M^{j+6-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) = 10a(n-1) - 22a(n-2) - 20a(n-3) + 63a(n-4) - 6a(n-5) - 24a(n-6)$$

for every $n > 5$.

Proof. The vector $w = q(M)\tau$ was formed by 6 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 5 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.